

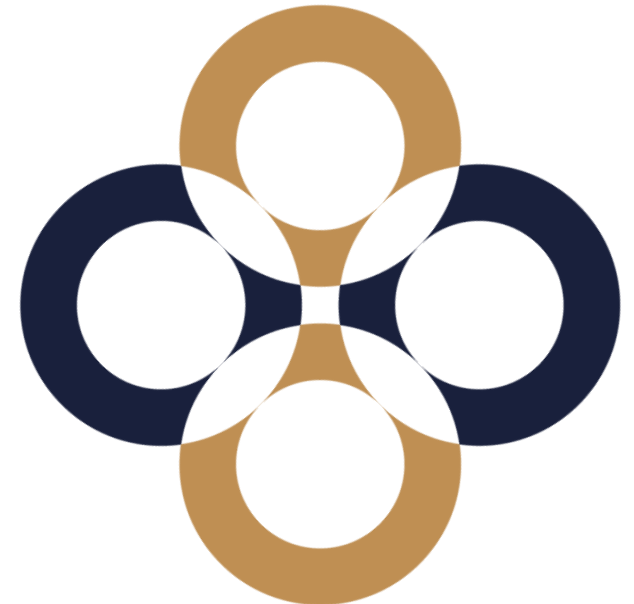
Capital Budgeting

International Corporate Finance – 7

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Based on:

Berk – DeMarzo (2017): Corporate Finance. Fourth edition, Pearson – chapter 8



Outline

Forecasting earnings

Determining free cash flows and NPV

Further adjustments to free cash flows

Analyzing the project

Forecasting earnings

- Capital Budget
 - Lists the investments that a company plans to undertake
- Capital Budgeting
 - Process used to analyze alternate investments and decide which ones to accept
- Incremental Earnings
 - The amount by which the firm's earnings are expected to change as a result of the investment decision

Forecasting earnings

■ Example

Linksys has completed a \$300,000 feasibility study to assess the attractiveness of a new product, HomeNet. The project has an estimated life of four years.

Revenue Estimates

Sales = 100,000 units/year

Per Unit Price = \$260

Forecasting Earnings

The \$7,500,000 is a cash expense and is not listed when calculating earnings

The firm uses straight-line **depreciation** for the new equipment

They will deduct depreciations equal to $\$7,500,000 / 5 \text{ years} = \$1,500,000$ per year

- Note, that only taxes are deducted when going from EBIT to net income
- Interest expense is typically not included in capital budgeting decisions
 - The project should be evaluated in its own right, not on how it will be financed

Forecasting Earnings

- Marginal Corporate Tax Rate

The tax rate on the marginal or **incremental** dollar of pre-tax income. Note: A negative tax is equal to a tax credit.

$$\text{Income Tax} = \text{EBIT} \times \tau_c$$

$$\begin{aligned}\text{Unlevered Net Income} &= \text{EBIT} \times (1 - \tau_c) \\ &= (\text{Revenues} - \text{Costs} - \text{Depreciation}) \times (1 - \tau_c)\end{aligned}$$

Forecasting Earnings

	Year	0	1	2	3	4	5
Incremental Earnings Forecast (\$000s)							
1 Sales	—	26,000	26,000	26,000	26,000	—	—
2 Cost of Goods Sold	—	(11,000)	(11,000)	(11,000)	(11,000)	(11,000)	—
3 Gross Profit	—	15,000	15,000	15,000	15,000	15,000	—
4 Selling, General, and Administrative	—	(2,800)	(2,800)	(2,800)	(2,800)	(2,800)	—
5 Research and Development	(15,000)	—	—	—	—	—	—
6 Depreciation	—	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7 EBIT	(15,000)	10,700	10,700	10,700	10,700	10,700	(1,500)
8 Income Tax at 40%	6,000	(4,280)	(4,280)	(4,280)	(4,280)	(4,280)	600
9 Unlevered Net Income	(9,000)	6,420	6,420	6,420	6,420	6,420	(900)

Taxing Losses for Projects in Profitable Companies

Problem

Kellogg Company plans to launch a new product. The new product will generate operating losses of \$15 million next year. Kellogg expects to earn pretax income of \$460 million from operations other than the new product next year. Kellogg's tax rate is 40%.

- What will Kellogg pay in taxes next year without the new product?
- What will Kellogg pay in taxes with the new product?

Forecasting Earnings

Solution:

Taxes without new product

Tax: $460 \times 40\% = 184$ million

Taxes with new product

Taxable income: $460 - 15 = 445$

Tax: $445 \times 40\% = 178$ million

New product reduces taxes by $184 - 178 = 6$ million

Forecasting Earnings

Opportunity costs presents an indirect cost of the value a resource could have provided in the best alternative use

For the HomeNet example, the new equipment will be housed in an existing lab, which cannot then be used for other purposes

If they could have rented it out for \$200,000 per year, then the after tax income from this would represent the opportunity costs

- The overhead is increased from \$2.800.000 to \$3.000.000.

Forecasting Earnings

- **Project Externalities** are indirect effects of the project that may affect the profits of other business activities of the firm.
- **Cannibalization** is when sales of a new product displaces sales of an existing product.
- In HomeNet example the sales of the existing router (price 100, cost 60) decreases by 25% (from 100,000)
 - $0.25 \times 100,000 \text{ units} \times \$100 \text{ per unit} = \$2,500,000 \text{ loss in sales}$
 - $0.25 \times 100,000 \text{ units} \times \$60 \text{ per unit} = \$1,500,000 \text{ reduction in costs}$

Forecasting Earnings

	Year	0	1	2	3	4	5
Incremental Earnings Forecast (\$000s)							
1 Sales	—	23,500	23,500	23,500	23,500	—	—
2 Cost of Goods Sold	—	(9,500)	(9,500)	(9,500)	(9,500)	—	—
3 Gross Profit	—	14,000	14,000	14,000	14,000	—	—
4 Selling, General, and Administrative	—	(3,000)	(3,000)	(3,000)	(3,000)	—	—
5 Research and Development	(15,000)	—	—	—	—	—	—
6 Depreciation	—	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7 EBIT	(15,000)	9,500	9,500	9,500	9,500	(1,500)	(1,500)
8 Income Tax at 40%	6,000	(3,800)	(3,800)	(3,800)	(3,800)	600	600
9 Unlevered Net Income	(9,000)	5,700	5,700	5,700	5,700	(900)	(900)

Forecasting Earnings

- **Sunk costs** are costs that have been or will be paid regardless of the decision whether or not the investment is undertaken.
 - Sunk costs should not be included in the incremental earnings analysis.
- **Fixed overhead expenses** should only be included if they are directly tied to the project, i.e. if the project requires additional expenses
- Past R&D expenditures are irrelevant (sunk costs), only incremental costs and benefits should be included

Forecasting Earnings

- Complications

Sales typically change every year

Prices also change

- Average costs per unit are typically not constant over time

Determining free cash flow and NPV

- The incremental effect of a project on a firm's available cash is its **free cash flow**.
- We are interested in the **free cash flows** generated by the project
We want to adjust the numbers for non-cash components
- **Capital expenditures** are actual cash flows and should be included when calculating free cash flows
- **Depreciation** is a non-cash expense and we need to adjust the free cash flows accordingly

Determining free cash flow and NPV

- We also need to adjust the **unlevered net income** to take into account the changes in **net working capital**

$$\begin{aligned}\text{Net Working Capital} &= \text{Current Assets} - \text{Current Liabilities} \\ &= \text{Cash} + \text{Inventory} + \text{Receivables} - \text{Payables}\end{aligned}$$

- **Trade credit** is the difference between receivables and payables.
- The change (increase) in **net working capital** is defined as
$$\Delta NWC_t = NWC_t - NWC_{t-1}$$

The HomeNet project is expected to have NWC equal to \$2,100,000 in year 1-4.

Determining free cash flow and NPV

		Year	0	1	2	3	4	5
Net Working Capital Forecast (\$000s)								
1	Cash Requirements		—	—	—	—	—	—
2	Inventory		—	—	—	—	—	—
3	Receivables (15% of Sales)		—	3,525	3,525	3,525	3,525	—
4	Payables (15% of COGS)		—	(1,425)	(1,425)	(1,425)	(1,425)	—
5	Net Working Capital		—	2,100	2,100	2,100	2,100	—

Net Working Capital

NWC is the capital required on the short term to run the business.
Main components are *cash, inventory, receivables, payables*.



Only that is needed to run the business on a day-to-day basis.

It doesn't include excess cash!!!



Can be invested at a market rate.

View as offsetting firm's debt → net debt...

Working capital (↓) can alter a firm's value (↑) by affecting its FCFF (↑).

Cash Cycle

The length of *time between* when a firm *pays cash to purchase its initial inventory* and when it *receives cash from the sale* of the output produced from that inventory.

Inventory on credit →
does not have to pay immediately.

Extend credits to customers →
delaying when to receive cash.

Cash Conversion Cycle (CCC) = Accounts Receivable Days + Inventory Days – Accounts Payable Days

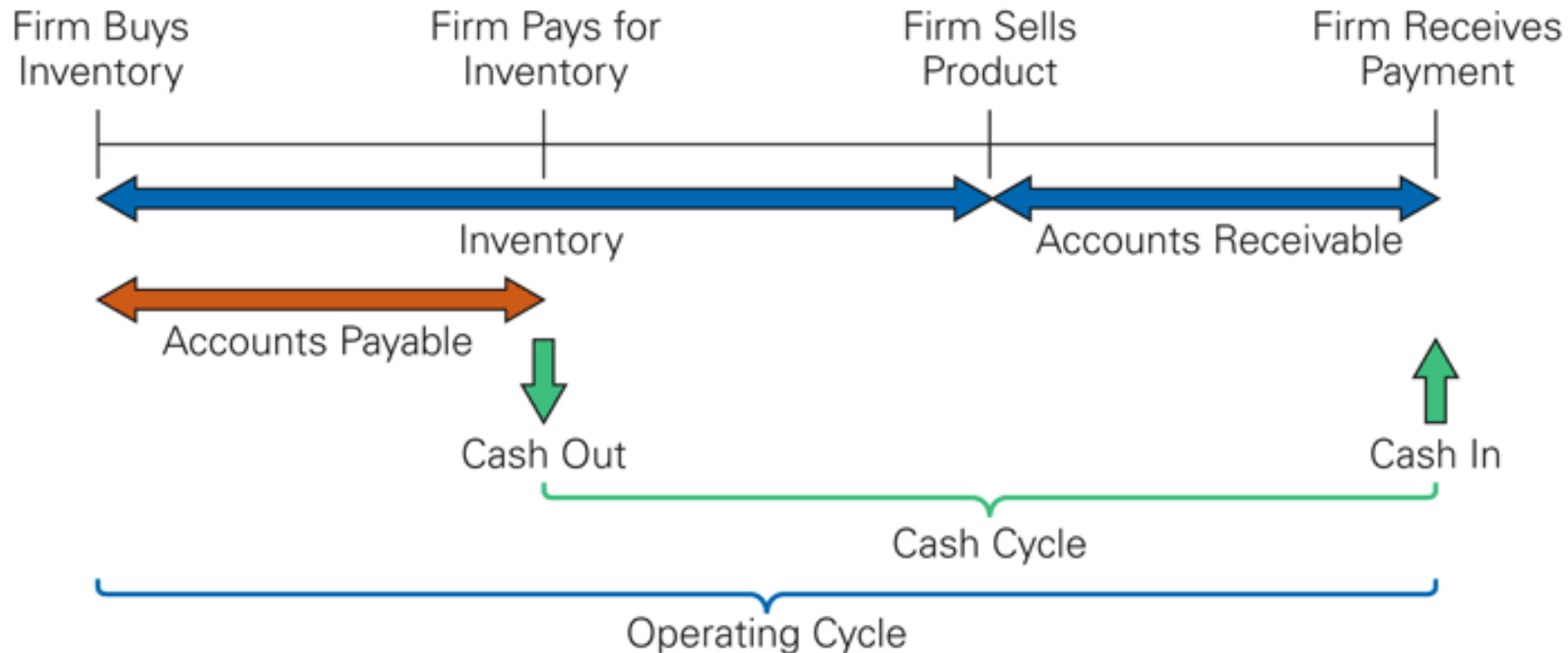
$$ARD = \frac{AR}{Sales/365}$$

$$ID = \frac{Inventory}{(Cost\ of\ Goods + SG\&A\ costs)/365}$$

$$APD = \frac{AP}{Cost\ of\ Goods / 365}$$

Operating Cycle

The average length of *time between* when a firm *originally purchases its inventory* and when it *receives the cash back* from selling its product.



Determining free cash flow and NPV

- Finally, we can define **free cash flows** as

$$\text{Free cash flows} = (\text{Revenues} - \text{Costs} - \text{Depreciation}) * (1 - \tau_c) + \text{Depreciation} - \text{CapEx} - \Delta NWC$$

- We can rewrite this and get

$$\text{Free cash flows} = (\text{Revenues} - \text{Costs}) * (1 - \tau_c) + \text{Depreciation} * \tau_c - \text{CapEx} - \Delta NWC$$

- The term $\tau_c * \text{Depreciation}$ is called the **depreciation tax shield**
- The **NPV** of a project is determined by discounting the free cash flows of the project

$$NPV = \sum_{t=0}^T \frac{FCF_t}{(1+r)^t}$$

Determining free cash flow and NPV

	Year	0	1	2	3	4	5
Incremental Earnings Forecast (\$000s)							
1 Sales		—	23,500	23,500	23,500	23,500	—
2 Cost of Goods Sold		—	(9,500)	(9,500)	(9,500)	(9,500)	—
3 Gross Profit		—	14,000	14,000	14,000	14,000	—
4 Selling, General, and Administrative		—	(3,000)	(3,000)	(3,000)	(3,000)	—
5 Research and Development		(15,000)	—	—	—	—	—
6 Depreciation		—	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7 EBIT		(15,000)	9,500	9,500	9,500	9,500	(1,500)
8 Income Tax at 40%		6,000	(3,800)	(3,800)	(3,800)	(3,800)	600
9 Unlevered Net Income		(9,000)	5,700	5,700	5,700	5,700	(900)
Free Cash Flow (\$000s)							
10 Plus: Depreciation		—	1,500	1,500	1,500	1,500	1,500
11 Less: Capital Expenditures		(7,500)	—	—	—	—	—
12 Less: Increases in NWC		—	(2,100)	—	—	—	2,100
13 Free Cash Flow		(16,500)	5,100	7,200	7,200	7,200	2,700

$$NPV = -16,500 + \frac{5,100}{1.12} + \frac{7,200}{(1.12)^2} + \frac{7,200}{(1.12)^3} + \frac{7,200}{(1.12)^4} + \frac{2,700}{(1.12)^5}$$

NPV=\$5,027 (000s)

Further Adjustments to Free Cash Flow

Liquidation or Salvage Value

- $\text{Capital Gain} = \text{Sale Price} - \text{Book Value}$
- $\text{Book Value} = \text{Purchase Price} - \text{Accumulated Depreciation}$
- $\text{After-Tax Cash Flow from Asset Sale} = \text{Sale Price} - (\tau_c \times \text{Capital Gain})$
- Suppose that in addition to the \$7.5 million in new equipment required for Home Net, some equipment will be transferred to the lab from another Cisco facility. This equipment has a resale value of \$2 million and a book value of \$1 million. If the equipment is kept rather than sold, its remaining book value can be depreciated next year. When the lab is shut down in year 5, the equipment will have a salvage value of \$800,000. What adjustments must we make to Home Net's free cash flow in this case?

Further Adjustments to Free Cash Flow

- Reduce Home Net's free cash flow in year 0 by the sale price less any taxes that would have been owed had the sale occurred: $\$2 \text{ million} - 40\% \times (\$2 \text{ million} - \$1 \text{ million}) = \1.6 million
- Year 1 depreciation tax shield $40\% \times \$1 \text{ million} = \$400,000$
- Year 5 after tax cash flow from sale: $\$800,000 \times (1 - 40\%) = \$480,000$

	Year	0	1	2	3	4	5
Free Cash Flow and NPV (\$000s)							
1	Free Cash Flow w/o equipment	(16,500)	5,100	7,200	7,200	7,200	2,700
	Adjustments for use of existing equipment						
2	After-Tax Salvage Value	(1,600)	—	—	—	—	480
3	Depreciation Tax Shield	—	400	—	—	—	—
4	Free Cash Flow with equipment	(18,100)	5,500	7,200	7,200	7,200	3,180
5	NPV at 12%	4,055					

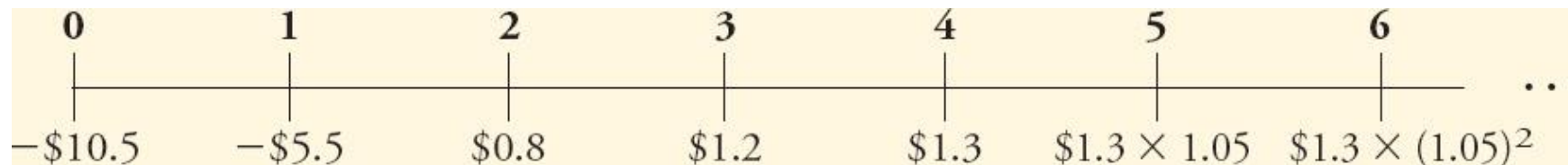
Further Adjustments to Free Cash Flow

- Terminal or Continuation Value

This amount represents the market value of the free cash flow from the project at all future dates.

Problem

Base Hardware is considering opening a set of new retail stores. The free cash flow projections for the new stores are shown below (in millions of dollars):



Constant growth after year 4: 5%

Cost of capital=10%

Terminal value? NPV?

Further Adjustments to Free Cash Flow

$$\begin{aligned}
 \text{Continuation Value in Year 4} &= PV(\text{FCF in Year 5 and Beyond}) \\
 &= \frac{FCF_4 \times (1+g)}{r-g} = \$1.30 \text{ million} \times \frac{1.05}{0.10 - 0.05} \\
 &= \$1.30 \text{ million} \times 21 = \$27.3 \text{ million}
 \end{aligned}$$

Year	0	1	2	3	4
Free cash Flow (years 0–4)	(10,500)	(5,500)	800	1,200	1,300
Continuation value					27,300
Free cash flow	(10,500)	(5,500)	800	1,200	28,600

$$NVP = -10,500 - \frac{5,500}{1.10} + \frac{800}{1.10^2} + \frac{1,200}{1.10^3} + \frac{28,600}{1.10^4} = \$5,567$$

Analyzing the Project

■ Break-Even Analysis

- The **break-even** level of an input is the level that causes the NPV of the investment to equal zero.
- HomeNet IRR Calculation

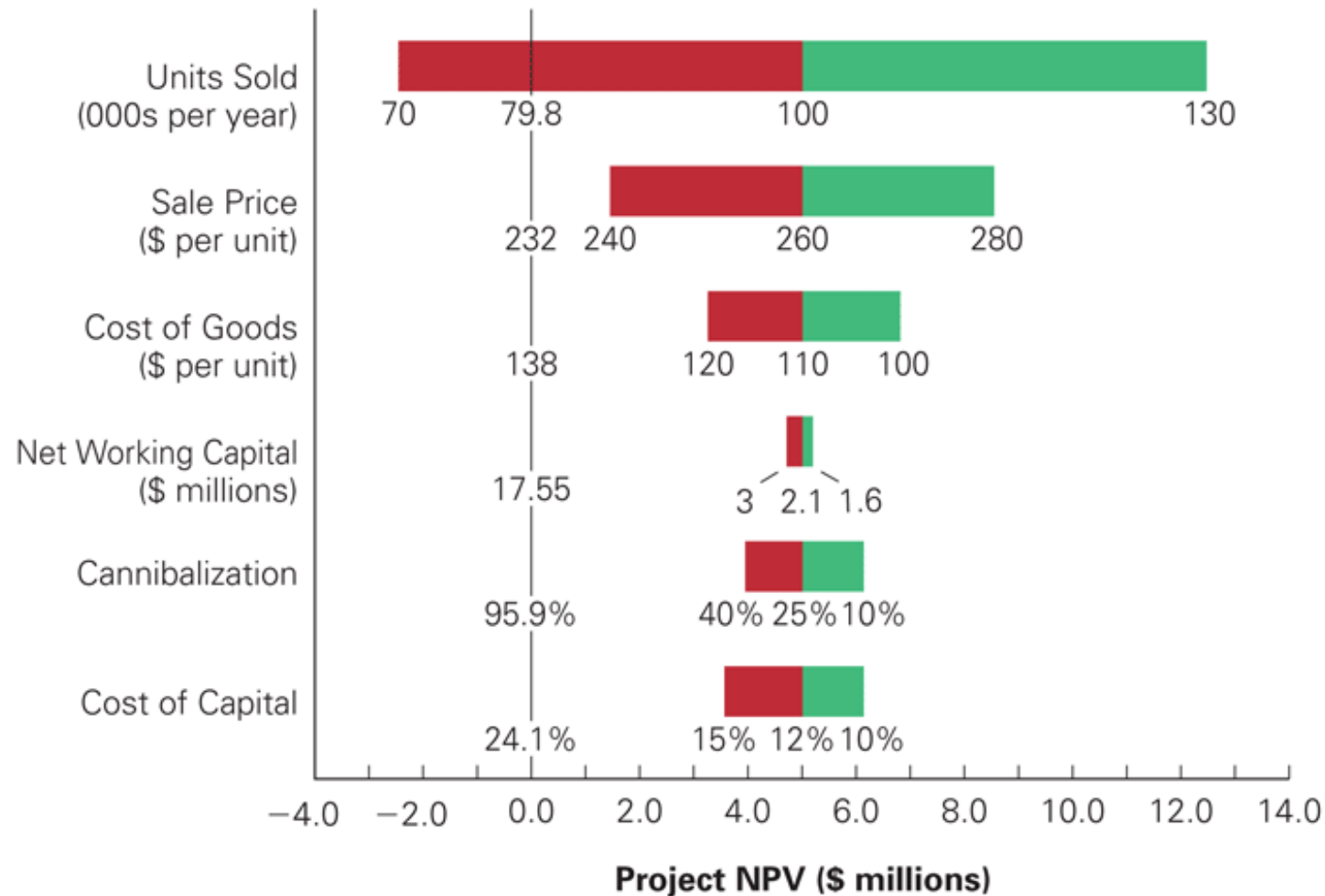
Year	0	1	2	3	4	5
NPV(\$000s)and IRR						
1 Free cash flow	(16,500)	5,100	7,200	7,200	7,200	2,700
2 NPV at 12%	5,027					
3 IRR	24.1%					
			Parameter		Break-Even Level	
			Units sold		79,759 units per year	
			Wholesale price		\$232 per unit	
			Cost of goods		\$138 per unit	
			Cost of capital		24.1%	

Analyzing the Project

- Sensitivity Analysis shows how the NPV varies with a change in one of the assumptions, holding the other assumptions constant.

Parameter	Initial Assumption	Worst Case	Best Case
Units sold (thousands)	100	70	130
Sale price(\$/unit)	260	240	280
Cost of goods(\$/unit)	110	120	100
NWC(\$ thousands)	2100	3000	1600
Cannibalization	25%	40%	10%
Cost of capital	12%	15%	10%

Sensitivity analysis





**Thank you
for your attention!**

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